

MERN Stack Practical Task

Duration: 4-5 Hours | **Candidate:** Nikita Dabhi | **Position:** MERN Stack Developer

Overview

Project: Job Portal with Authentication
Build a job portal application where employers can post jobs and candidates can apply.

Requirements

Backend (Node.js + Express + MongoDB)

1. Database Schema Design

- Users:** `_id`, name, email, password (hashed), role (employer/candidate), createdAt
- Jobs:** `_id`, title, company, description, location, salary, skills[], postedBy (ref User), createdAt
- Applications:** `_id`, jobId (ref Job), candidateId (ref User), resume, status (pending/accepted/rejected), createdAt

2. REST API Endpoints

Method	Endpoint	Description
POST	/api/auth/register	Register user (with role selection)
POST	/api/auth/login	Login, return JWT
POST	/api/jobs	Create job (employer only)
GET	/api/jobs	Get all jobs (with filters)
	/api/jobs/:id	Get job details
PUT	/api/jobs/:id	Update job (owner only)
DELETE	/api/jobs/:id	Delete job (owner only)
POST	/api/applications	Apply for job (candidate only)
GET	/api/applications/my	Get my applications
GET	/api/applications/job/:jobId	Get job applications (employer)

Method	Endpoint	Description
PATCH	/api/applications/:id	Update status (employer)

3. Features

- JWT authentication with role-based access
- **Employer:** Can create, update, delete jobs
- **Candidate:** Can apply to jobs, view applications
- Search jobs by title, location, skills
- Filter by salary range

Frontend (React)

1. Pages & Components

- **Login Page**
- **Register Page** (with role selection)
- **Home Page** (job listings with search)
- **Job Detail Page**
- **Employer Dashboard** (My Jobs, Applications received)
- **Candidate Dashboard** (Applied Jobs)
- **Post Job Form** (employer only)

2. Features

- JWT token stored in localStorage
- Protected routes based on role
- Search functionality & Responsive design
- Loading and error states

Guidelines

Backend Technical Guidelines

- **Security:** Use bcrypt for password hashing and jsonwebtoken for JWT
- **Database:** Use MongoDB
- **Logic:** Proper error handling with try-catch and input validation

Deliverables

- GitHub repository with frontend and backend
- **README** with setup instructions and environment variables
- Postman collection for API testing

Environment Variables

None

PORT=5000

MONGODB_URI=mongodb://localhost:27017/job-portal

JWT_SECRET=your-secret-key

JWT_EXPIRE=7d

Submission

- Push code to GitHub and share the repository link
- Include setup instructions in README

Expected completion time: Today 7:00 PM - 07-Aug-2026

Maximum submission time: 7:00 PM - 08-Aug-2026

Good Luck!